

PUBLIC SUBMISSION

As of: March 21, 2025
Received: March 15, 2025
Status: [REDACTED]
Tracking No. m8a-jgyb-krlu
Comments Due: March 15, 2025
Submission Type: API

Docket: NSF_FRDOC_0001
Recently Posted NSF Rules and Notices.

Comment On: NSF_FRDOC_0001-3479
Request for Information: Development of an Artificial Intelligence Action Plan

Document: NSF_FRDOC_0001-DRAFT-2282
Comment on FR Doc # 2025-02305

Submitter Information

Name: Joshua Levine
Email: [REDACTED]

General Comment

See attached file(s)

Attachments

AI RFI Levine



Comment on Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

Joshua Levine

Research Fellow, Foundation for American Innovation.

March 15, 2025

Thank you for the opportunity to comment on the National Science Foundation's (NSF) and the Network and Information Technology Research Development's (NITRD) request for information on the development of the Artificial Intelligence (AI) Action Plan. In my comment, I seek to draw attention to an issue that significantly impacts AI development: copyright law. This comment is adapted from a recent paper I co-authored with Tim Hwang, titled: "Copyright AI, and Great Power Competition."¹

This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.

1. Introduction

Artificial intelligence (AI) has the makings of a general-purpose technology (GPT)

¹ Joshua Levine, Tim Hwang, "Copyright, AI, and Great Power Competition," Foundation for American Innovation, (January, 2025), <https://www.thefai.org/posts/copyright-ai-and-great-power-competition>.

that could usher in a new industrial revolution.² As a GPT, AI can be diffused into existing processes while creating new products, services, and markets. This has significant implications for national soft power (economic strength, technological adoption, cultural impact) and hard power (military capabilities, statecraft, national sovereignty, and geopolitical competition).

The most significant geopolitical AI competition is between the United States and China. The Chinese Communist Party (CCP) has made clear its desire to change the status quo, with AI featuring heavily in such plans.³ In 2018, President Xi Jinping stated that China must "ensure that our country marches in the front ranks where it comes to theoretical research in this important area of AI, and occupies the high ground in critical and AI core technologies."⁴ In 2023, Xi directed the People's Liberation Army to "raise the presence of combat forces in new domains and of new qualities."⁵

2. Copyright Law as a Threat to U.S. AI Leadership

There is one area of law that could deal a serious blow to U.S. efforts to maintain AI leadership: copyright. Nearly all leading AI model developers in the U.S. face lawsuits from copyright holders claiming that the inclusion of copyrighted material in training materials constitutes infringement. Currently, 39 open copyright cases against model developers seek damages and ask for data sets to be destroyed and models retrained. If plaintiffs are successful, they would cause irreparable harm to

² Nicholas Craft, "Artificial Intelligence as a General-Purpose Technology: An Historical Perspective," *Oxford Review of Economic Policy*, vol. 37, no. 3 (September 23, 2021), p. 521, <https://academic.oup.com/oxrep/article/37/3/521/6374675>.

³ Arthur R. Kroeber, "Unleashing 'New Quality Productive Forces': China's Strategy for Technology-Led Growth," *Brookings Institution*, June 4, 2024, <https://www.brookings.edu/articles/unleashing-new-quality-productiveforces-chinas-strategy-for-technology-led-growth/>.

⁴ Elisa Kania, Roger Creemers, "Xi Jinping Calls for 'Healthy Development' of AI (Translation)," *New America*, November 5, 2018, <https://www.newamerica.org/cybersecurity-initiative/digichina/blog/xi-jinping-calls-forhealthy-development-of-ai-translation/>.

⁵ Jacob Stokes, "Military Artificial Intelligence, the People's Liberation Army, and U.S.-China Strategic Competition," *Center for New American Security*, February 1, 2024, <https://www.cnas.org/publications/congressional-testimony/military-artificial-intelligence-the-peoples-liberation-army-and-u-s-china-strategic-competition>

America's AI sector, with grave implications for America's economy, military capabilities, and geopolitical standing. Identifying the threat copyright lawsuits pose to American model developers and operationalizing policies to mitigate potential harms will be critical to advancing American competitiveness in AI development.

3. The Case for Fair Use in AI Training

Current "frontier" AI models would not exist without the ability to access and analyze vast amounts of data available online. Training large-scale models using prevailing scaling techniques requires massive data sets that can provide the necessary diversity, complexity, and quality of data.⁶ The default rule under U.S. copyright law is that unauthorized copying of material is actionable infringement. However, AI firms argue that accessing and analyzing such data constitutes fair use.

Fair use is a doctrine intertwined with copyright law for centuries and officially enshrined in the Copyright Act of 1976.⁷ It allows for nonpermissive use of copyrighted works in a transformative manner as a counterbalance to the limited monopoly that copyright holders enjoy. It is seen as a counterbalance to the limited monopoly that copyright-holders enjoy over their work to "promote the Progress of Science and useful Arts," the original justification for copyright protection.⁸ Fair use is determined on a case-by-case basis, with judges relying on a four-part test to determine whether the use of a copyrighted work qualifies as fair use.⁹ The four aspects considered are:

1. The purpose and character of the use, including whether such use is of a commercial nature or for nonprofit educational purposes;
2. The nature of the copyrighted work;
3. The amount and substantiality of the portion used in relation to the

⁶ Lukas Budach et al., "The Effects of Data Quality on Machine Learning Performance," ArXiv, (November 9, 2022), <https://arxiv.org/abs/2207.14529>.

⁷ Copyright Act of 1976, 17 U.S.C. § 107 (1976).

⁸ U.S. Const. art. I, § 8, cl. 8.

⁹ Richard Stim, "Measuring Fair Use: The Four Factors," Stanford University Libraries, accessed November 5, 2024, <https://fairuse.stanford.edu/overview/fair-use/four-factors/>.

copyrighted work as a whole; and

4. The effect of the use upon the potential market for or value of the copyrighted work.

The plaintiffs in the various lawsuits are a mix of creatives, such as comedians and artists, and corporate rights-holders, such as the New York Times and Getty Images.¹⁰ While each complaint is unique, all the cases assert that the use of copyrighted works without permission or payment is a willful infringement of copyright law, and that model developers must pay significant fines, destroy any models that were trained on such materials, and commit to not using copyrighted data in training in the future.

It is worth noting that this response is not new in some respects. Responses to new technologies are often severe, especially when incumbents and politically connected groups face disruptive or uncertain futures.¹¹ In the early 2000s, file-sharing platform Napster was sued by record labels for its role in the distribution of pirated music.¹² This litigation ultimately bankrupted the company and halted the use of file-sharing broadly. Not content with disrupting the use of the technology alone, the Recording Industry Association of America began suing individuals who shared or downloaded music outside official channels.¹³ Distortion, fear mongering, and attempts to halt innovation to preserve their place in the market have been hallmarks of such incumbents, making a clear-eyed and nuanced analysis all the more critical in the case of AI training and copyright.¹⁴

There are strong arguments for protecting under fair use the use of copyrighted material in training, and a line of jurisprudence stretching back more than 50

¹⁰ “Master List of Lawsuits v. AI, ChatGPT, OpenAI, Microsoft, Meta, Midjourney & Other AI Cos,” ChatGPT is Eating the World, accessed on March 14, 2025, <https://chatgptiseatingtheworld.com/2024/08/27/master-list-oflawsuits-v-ai-chatgpt-openai-microsoft-meta-midjourney-other-ai-cos/>.

¹¹ Pamela Samuelson, “Fair Use Defenses in Disruptive Technology Cases,” UCLA Law Review, Forthcoming, November 28, 2023, https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4631726.

¹² A&M Records, Inc. v. Napster, Inc., No. 239 F.3d 1004 (9th Cir. 2001), <https://www.copyright.gov/fair-use/summaries/a&mrecords-napster-9thcir2001.pdf>.

¹³ David Kravets, “Copyright Lawsuits Plummet in Aftermath of RIAA Campaign,” Wired, May 18, 2010, <https://www.wired.com/2010/05/riaa-bump/>.

¹⁴ See Louis Anslow, “Pessimists Archive,” accessed Dec. 5, 2024, <https://pessimistsarchive.org/>, especially for “novels,” “radio,” and “television.”

years can be instructive on the questions being raised today.

Three key themes from successful fair use defenses apply to AI training:

1. Fair use allows people to share information, helping to uncover new insights and uses.¹⁵
2. Fair use protects the right to learn and iterate upon prevailing ideas.¹⁶
3. Fair use ensures a counterweight to the monopoly on ideas that copyright creates.¹⁷

The common threads among these cases are the expansion of access to information, an opportunity to create new products or services, and the freedom to learn and build upon existing knowledge to create anew. When considering the promises of AI diffusion and improvement and how the law should treat this budding technology, these ideas should be front of mind.

Current copyright lawsuits, however, threaten such promises and risk stunting AI progress in the United States and boosting the standing and competitiveness of the Chinese AI apparatus.

4. Costs if U.S. Falls Behind in AI Development

4.1 Direct Costs from Litigation

Copyright infringement fines can range from \$750 to \$30,000 per infringement, with maximum penalties of \$150,000 for willful infringement.¹⁸ Given the size of AI training datasets, successful lawsuits could result in fines totaling billions or

¹⁵ Pamela Samuelson, “Fair Use Defenses in Disruptive Technology Cases,” UCLA Law Review, Forthcoming, November 28, 2023, https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4631726; Authors Guild, Inc. v. Google Inc., No. 13-4829-cv (2d Cir. Oct. 16, 2015), <https://www.copyright.gov/fair-use/summaries/authorsguild-google-2dcir2015.pdf>.

¹⁶ Sega Enters. Ltd. v. Accolade, Inc., 977 F.2d 1510 (9th Cir. 1992), <https://copyright.gov/fair-use/summaries/segaenters-accolade-9thcir1992.pdf>.

¹⁷ Sony Corp. of Am. v. Universal City Studios, Inc., 464 U.S. 417 (1984), <https://www.copyright.gov/fair-use/summaries/sonycorp-universal-1984.pdf>.

¹⁸ Copyright Act of 1976, 17 U.S.C. § 107, ch. 5 (1976).

even trillions of dollars, potentially bankrupting leading AI model developers.¹⁹ Beyond the monetary fines, some plaintiffs are asking courts to require the re-training or destruction of AI models whose training data included copyrighted works.²⁰

4.2 Ripple Effects Across the AI Stack

The second-order effects of adverse judgments would impact the entire AI stack, including:

- **Hardware:** Reduced demand for high-performance semiconductors and custom computing hardware, undermining U.S. investments in semiconductor production.²¹
- **Data Centers:** Diminished investment in data center infrastructure, which currently exceeds \$400 billion annually.²²
- **Applications:** Stifled innovation in AI applications and startups. In the private sector, leading startup accelerator Y Combinator has reported that more than seventy-five percent of its summer cohort were focusing on AI models and tools, and fifty percent of its most recent cohort relied on AI

¹⁹ Pamela Samuelson, “How to Think About Remedies in the Generative AI Copyright Case,” Lawfare, February 15, 2024, <https://www.lawfaremedia.org/article/how-to-think-about-remedies-in-the-generative-ai-copyright-cases>; Joshua Levine, “To Support AI, Defend the Open Internet and Fair Use,” Tech Policy Press, October 17, 2024, <https://www.techpolicy.press/to-support-ai-defend-the-open-internet-and-fair-use/>.

²⁰ Getty Images (US), Inc. v. Stability AI, Inc., 1:23-cv-00135, (D. Del.); The New York Times Company v. Microsoft Corporation, OpenAI, Inc., 1:23-cv-11195, (S.D. NY); “Books3 AI Training Dataset.” AIAAIC, last updated October 2024, <https://www.aiaaic.org/aiaaic-repository/ai-algorithmic-and-automation-incidents/books3-ai-trainingdataset>.

²¹ Dick Weisinger, “Custom Silicon: How Tech Companies are Designing Their Own Chips,” Formtek, December 19, 2023, <https://formtek.com/blog/custom-silicon-how-tech-companies-are-designing-their-own-chips/>; “FACT SHEET: CHIPS and Science Act Will Lower Costs, Create Jobs, Strengthen Supply Chains, and Counter China,” The White House, August 9, 2022, <https://www.whitehouse.gov/briefing-room/statementsreleases/2022/08/09/fact-sheet-chips-and-science-act-will-lower-costs-create-jobs-strengthen-supply-chainsand-counter-china/>.

²² Matt Ashare, “Data Center Spending Skyrockets as Cloud Building Rush Accelerates,” CIO Dive, September 30, 2024, <https://www.ciodive.com/news/cloud-data-center-hyperscale-spending-spree-generative-ai/728469/>

models and tools to support their operations.²³

4.3 Military and National Security Implications

The Department of Defense (DOD) already procures systems from commercial firms or open-source models and fine-tunes them for military capabilities.²⁴ Specifically, models from Meta and OpenAI are being integrated and deployed to support American military and intelligence operations.²⁵ Legal restrictions on AI development would inhibit the government from accessing and deploying the private sector's most advanced models, putting American military and intelligence communities at a disadvantage compared to Chinese counterparts.

4.4 Opportunity Costs and Free Expression Impact

Beyond direct investment losses, there are significant opportunity costs in fields poised for AI-driven efficiency gains, such as drug development, physical infrastructure, and transportation. Copyright maximalism could also restrict the use of AI tools for individual expression, criticism, and speech.

5. How Copyright Restrictions Benefit China

Chinese model developers operate under two sets of rules: strict regulations for "public use" within China, and more permissive conditions for research,

²³ William Coulman and David Crowther, "Three-Quarters of Founders in the Latest Y Combinator Cohort Are Working on AI Startups," Sherwood, August 23, 2024, <https://sherwood.news/tech/three-quarters-of-founders-in-the-latest-y-combinator-cohort-are-working-on/>.

²⁴ Defense Innovation Board, Department of Defense, Aligning Incentives to Drive Faster Tech Adoption (2024), https://innovation.defense.gov/Portals/63/20240701%20DIB%20Report_Aligning%20Incentives%20PUBLISHED%20STUDY_1.pdf.

²⁵ Brandi Vincent, "Scale AI Unveils 'Defense Llama' Large Language Model for National Security Users," DefenseScoop, November 4, 2024, <https://defensescoop.com/2024/11/04/scale-ai-unveils-defense-llama-large-language-model-llm-national-security-users/>; "Anduril Partners with OpenAI to Advance U.S. Artificial Intelligence Leadership and Protect U.S. and Allied Forces," Anduril Industries, December 4, 2024, <https://www.anduril.com/article/anduril-partners-with-openai-to-advance-u-s-artificial-intelligence-leadership-and-protect-u-s/>.

enterprise, or industrial uses.²⁶ This dual approach allows China to control domestic information while enabling Chinese firms to compete globally.

China has exempted model developers from data and content restrictions, including copyright limitations, if the model is designed for "enterprise, education and research, or industry association." This gives Chinese developers significant advantages in leveraging vast amounts of data and an incentive to partner with industry and research organizations to protect investment and scale the technology.

Chinese model development is and will continue to support the CCP's strategic and military might. The CCP's expansive surveillance capabilities are being enhanced with AI to monitor dissent.²⁷ Chinese firms are producing innovative products combining AI models with data to improve sensors, radar systems, and computer vision—all dual-use technologies with both civilian and military applications.²⁸

If the United States loses its leadership position in AI, China will fill the void, with negative consequences domestically and globally. Unlike in China, American AI developers do not have to abide by government-imposed speech codes or socialist values, and they can choose whether to partner with the state. Allowing Chinese AI models to be developed and diffuse more quickly will harm both American soft and hard-power.

6. Policy Recommendations

The legal battles around AI and copyright raise the specter that a domestic, commercial squabble of interests might prevent the United States from retaining

²⁶ Yi Wu, "How to Interpret China's First Effort to Regulate Generative AI Measures," China Briefing, July 27, 2023, <https://www.china-briefing.com/news/how-to-interpret-chinas-first-effort-to-regulate-generative-ai-measures/>.

²⁷ Mara Hvistendahl, "How China Surveils the World," MIT Technology Review, August 19, 2020, <https://www.technologyreview.com/2020/08/19/1006455/gtcom-samantha-hoffman-tiktok/>.

²⁸ Elisa B. Kania, Testimony before the U.S.-China Economic and Security Review Commission Hearing on Trade, Technology, and Military-Civil Fusion, June 7, 2019, https://s3.us-east-1.amazonaws.com/files.cnas.org/backgrounds/documents/June-7-Hearing_Panel-1_Elsa-Kania_Chinese-Military-Innovation-in-ArtificialIntelligence.pdf?mtime=20190617115242&focal=none.

its edge in the global competition around the technology. The victory of the litigants in the many pending lawsuits against AI companies may lead to legal regimes that ultimately impose ruinous monetary damages, allow the extraction of burdensome settlements, and in extreme cases result in the court-ordered destruction of AI models themselves. All would do great harm to the emerging market of products and services leveraging the technology.

Regardless of the outcome, the persistent uncertainty around the legal standing of training raises real costs, as well. In all likelihood, pending lawsuits that appear likely to set precedent in this area will take years to resolve. Facing this uncertainty, leading AI developers may very well choose to rebase their model development operations in countries with more favorable environments for training. Other nations, such as Japan and Singapore, have explicitly carved out exceptions for the training of AI systems, and others could follow suit in an effort to capitalize on a sustained period of American inaction.

The state of the research field and the market for AI services is evolving fast enough that the United States cannot afford to wait on the slow process of judicial policymaking in this area. Properly constituted, copyright can serve not as a threat to technological progress and American advantage, but as an asset. I propose four immediate policy actions to this end.

6.1 Congressional Clarification of Fair Use

Congress has the power to settle the copyright status of training immediately. It should pass a bill that asserts unequivocally that all forms of training AI systems are a fair use, providing these activities with a clear mandate to move forward. Such a definitive approach would:

- Immediately lower the risk of engaging in model training, encouraging new entrants to join the market
- Counter the existential risk posed by pending lawsuits against leading American AI companies
- Allow American AI companies to continue growing toward global dominance

6.2 Secure Training Data for American Industry

Training data—particularly high-quality training data—is an increasingly scarce commodity even as it remains critical for advancing AI capabilities.²⁹ Global competitiveness depends not just on access to talent and advanced semiconductors, but also on securing the best datasets.

The U.S. Trade Representative should prioritize bringing the full weight of American commercial power to bear on opening up and securing valuable sources of training data worldwide for U.S. firms. This may include:

- Trade deals that carve out fair use-style exceptions for model training in the intellectual property frameworks of other nations
- Agreements to obtain exclusive access to valuable datasets in other countries, including multilingual materials, scientific corpora, and multimodal datasets

6.3 Establish the U.S. Bureau of Data Management

Some of the most valuable untapped datasets for AI advancement exist within the government. The formation of a U.S. Bureau of Data Management (BDM) within the Department of Commerce could accelerate U.S. leadership in AI.

The BDM would serve as a dedicated team of regulatory experts and technologists tasked with:

- Identifying and "liberating" valuable government datasets that could be open-sourced or licensed to AI developers
- Serving as a clearinghouse for archival collections, legal codes, weather data, and other valuable information
- Gathering feedback from industry about the most valuable datasets to prioritize
- Working with state and local governments to make their data available for AI training

6.4 Accelerate Opt-Out Standards

²⁹ Ilya Sutskever, "Sequence to Sequence: Learning with Neural Networks: What a Decade," NeurIPS, December 13, 2024, <https://www.youtube.com/watch?v=1yvBqasHLZs>.

Copyright is being used as a lever for parties seeking to defend the right of individuals to "opt-out" of having their creative work used in AI training. While this is an important right to preserve, litigation is not the best method to achieve this end.

Clear technical protocols and strong norms can help foster an ecosystem where opting out is easily indicated and respected. "Robots.txt," an informal voluntary standard that indicates whether a website can be indexed by bots, demonstrates this can work at scale.³⁰

The government should establish a task force bringing together technology companies, academic researchers, rights-holders, and technical experts to establish and promote a common standard allowing creators to clearly indicate when copying for AI training is permitted or not.

7. Conclusion

Debates around copyright and intellectual property have existed since America's founding, evolving to balance incentives for creation with access to information to promote progress. AI's emergence has added a new dimension to these questions, as it represents a general-purpose technology with the potential to reshape economies, militaries, and societies.

The United States is positioned to lead in AI development and deployment, but China is actively working to surpass American capabilities. Copyright lawsuits currently present the greatest impediment to American AI progress. To ensure continued U.S. leadership, clarity around accessing data and training models under the fair use doctrine is paramount.

The policy recommendations outlined above would provide immediate relief to the AI industry while securing America's competitive advantage in this critical technology domain. By taking these steps, the United States can maintain its technological edge, strengthen national security, and promote economic prosperity in the AI era.

³⁰Jo Cameron, "What is a Robots.txt File? Best Practices Robot.txt Syntax," Moz, updated November 7, 2024, <https://moz.com/learn/seo/robotstxt>.